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Immune or At-Risk? Stock Markets and the Significance of the COVID-19 Pandemic

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Abstract:

Quantifying influential price drivers in times of economic uncertainty will perpetually occupy the work of academics and stock market practitioners alike. Recently, the unprecedented closure of borders, workplaces, and traditional commerce due to the COVID-19 pandemic has had significant financial impacts. Examining stock markets in five global epicenters of the virus, this study implements stepwise linear regression with the Bayesian Information Criterion to identify whether growth in the virus was significantly associated with equity index prices. In this study, the daily total number of confirmed COVID-19 cases was significantly associated with index prices in Spain (IBEX-35), Italy (FTSE MIB), the U.K. (FTSE 100) and the U.S.A. (DJIA) during the pandemic period spanning 31st December 2019 to 10th June 2020. However, similar results were not observed for China (SSE 180), the global origin of the virus. Research emerging from China by He et al. (2020) suggests a significant underreporting of the pandemic, indicating a potentially spurious dataset currently available for analysis. Excluding China, these findings concur with the emerging research on the impact of COVID-19 on financial markets, confirming a negative and significant relationship between the growth in COVID-19 and major equity index prices. Moreover, this study adds to the previous literature by controlling for volatility, monetary policy, counterparty credit risk, liquidity risk, safe-haven asset demand and the price of oil, using a continuous and temporal flow of data. As such, this study provides a robust set of findings on this contemporary matter.

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1. Introduction

1.1. Background

Originating in China in late 2019 and declared as a global pandemic by the WHO on March 12th 2020, the current outbreak of COVID-19 has been arguably the greatest health crisis of the 21st century, prompting travel restrictions, stay-at-home orders and an unprecedented shutdown of global commerce. As of June 19th, 2020, global cumulative cases stood at 8,618,777 with 457,269 recorded deaths, amounting to a fatality rate of approximately 9% (ECDC 2020). From Table 1 it can be seen that although the virus is not significantly more fatal than preceding viral outbreaks, the transmission rate of COVID-19 is currently estimated to be substantially higher.

With this higher rate of transmission, the economic impact, limitedly observable in the short-term through the stock market, has also been notably more severe relative to past outbreaks. On March 12th, 2020, both the Dow Jones and S&P 500 experienced their worst trading days since Black Monday of 1987, falling over 9% and subsequently hitting 52-week lows (Stevens et al. 2020). However, large stock market losses were accompanied by similarly large upward movements throughout March, as markets reacted turbulently to developments in the outbreak, among other market factors. One such factor was the tension in global oil markets between Saudi Arabia and Russia, which may have propagated the distress of an already volatile market. Table 2 illustrates the reaction of the S&P 500 and the peak-to-trough price movement of the index during the outbreaks mentioned previously, highlighting COVID-19 as by far the most significant outbreak to affect financial markets this century.

Table 1: *Fatality and Transmission Rate of Previous Viral Outbreaks*

Table 1: Presents both the fatality and transmission rates of major viral outbreaks. The fatality rate of COVID-19 is not significantly higher than previous outbreaks. However, current estimates of the transmission rate (R_0) indicate as many as 5.7 cases generated for each new case.

Year	Outbreak	Fatality Rate	R_0 (Transmission Rate)
(2003)	SARS [1]	11%	3.0 [1]
(2009)	Swine Flu [2]	1%	1.6 [7]
(2012)	MERS [3]	35%	0.5 [8]
(2014)	Ebola Virus [4]	50%	1.7 [9]
(2016)	Zika Virus [5]	8%	3.8 [10]
(2020)	COVID-19 [6]	9%	5.7 [11]

Sources: [1] WHO 2003; [2] Dawood et al. 2012; [3] WHO 2019; [4] WHO 2020; [5] Cunha et al. 2017; [6] ECDC 2020 [7] Coburn et al. 2009, [8] Poletto 2014, [9] Khan et al. 2015 [10] Towers et al. 2016, [11] Sanche et al. 2020.

Table 2: *S&P 500 Peak-to-Trough Returns During Previous Viral Outbreaks*

Table 2: Presents S&P 500 Index price movements during major viral outbreaks. Peak-to-trough losses were almost two-fold during COVID-19 when compared to SARS. Notably, taking just 23 trading days versus 38 during the SARS virus.

Year	Outbreak	Peak-to-Trough Returns	Peak-to-Trough Trading Days
(2003)	SARS [12]	-17.3%	38
(2009)	Swine Flu [13]	-5.8%	9
(2012)	MERS [12]	-7.3%	43
(2014)	Ebola Virus [12]	-5.8%	23
(2016)	Zika Virus [12]	-12.9%	66
(2020)	COVID-19 [13]	-33.7%	23

Sources: [13] Citi Market Insights 2020; [14] Bloomberg Terminal 2020

The true economic impact of COVID-19 will remain unseen for some time. At the time of writing, countries are beginning to ease lockdown restrictions as people slowly return to work. With this, countries are only now beginning to take stock of the economic, social, and political costs of both the pandemic and the subsequent economic shutdown. A report by the Bank of England (2020) estimated a G.D.P. contraction of 25% in the June quarter with an estimated contraction of 14% by year-end. Similar estimates of 20%-30% were announced by the Chairman of the Federal Reserve, Powell (2020), alongside predictions from the I.M.F. (2020) that the COVID-19 recession will likely be the worst since the Great Depression. However, before the long-term economic impacts can be quantified, an analysis into the short-term stock market implications is warranted. Global stock market losses of $16 trillion were observed in less than one month as the pandemic took hold and as fears rose of a worldwide recession (Gandel 2020). As such, an analysis into the determinants of global stock market indices as COVID-19 took hold will provide valuable insights into the evolving market dynamics and price drivers during times of crisis and uncertainty.

Regression analysis techniques have long been used in literature to assess the financial impact of distressful events, ranging from banking crises to terrorist attacks and natural disasters (Jiang et al. 2017; Tavor and Teitler-Regev 2019; Bergholt and Lujala 2012). In this study, using multiple regression and stepwise variable selection, the relationship between growth in COVID-19 and stock market index prices is investigated. Recent literature on the topic, detailed in Section 1.3, has found a negative and significant relationship between the growth in COVID-19 and stock market returns. Using the total number of COVID-19 cases and a selection of theoretical market determinants, detailed in Section 2.2, this analysis uses stepwise regression techniques to identify whether growth in COVID-19 itself was significantly associated with index prices.

1.2. Research Motivation & Findings

Despite the potential for viral outbreaks to affect financial markets, previous literature on the topic is scant. Namely, there is a lack of research on the short-term reaction of stock markets to the onset of a viral outbreak. Recent research investigating the tail risk of contagious diseases by Cirillo and Taleb (2020) has identified the distribution of fatalities in these events as strongly fat-tailed, suggesting the presence of a tail risk that is largely ignored in common epidemiological models. Additionally, a research note published as early as January 26th (2020) by Norman, Bar-Yam and Taleb warned of the systemic risk of a pandemic and the risk of societal ruin, should adequate responses not be taken. It would be more than a month after this publication before countries finally implemented the responsive measures it recommended. Additionally, recent literature has described the COVID-19 pandemic as a stock market black swan event (Morales and Andreosso-O'Callaghan 2020). However, it is evident that viral outbreaks are inevitable in society. In hindsight, it can also be seen that the dangers posed by novel pathogens have in fact been highlighted for many years (Garrett 1994, Gates 2015; Norman et al. 2020). The danger of treating the COVID-19 pandemic as a black swan is that there may be a subsequent failure to adequately prepare for the next pandemic, since the rareness and exceptional nature of such events can be used to reject the likelihood and severity of future reoccurrences. It also detracts focus from the necessary risk analysis and predictive modelling that should be carried out following such events. Goodell (2020) explored agendas for future COVID-19 research, arguing that it is very likely the next time there is a sudden and significant viral outbreak, there will concomitantly be a substantial reaction from financial markets. Travel restrictions, social distancing measures and temporary shutdowns of commerce are expected to be the standard course of action when a novel pathogen inevitably emerges once more (Khanna et al. 2020; Bearman et al. 2020). Therefore, it is imperative that the economic and financial impact of pandemics is further investigated.

This study uses the limited COVID-19 data available from the ongoing pandemic to analyze the relationship between COVID-19 and stock market indices, encompassing the period from the initial growth of COVID-19 to the relative flattening of the curve. As an outbreak unfolds, relevant data is negligible, and this study analyses whether the total number of COVID-19 cases was significant in driving stock market indices and the immediate reaction of investors. This study adds to the current literature in several ways. Firstly, recently published research on this topic has observed a negative relationship between COVID-19 related cases and stock market returns (Ashraf 2020; Al-Awadhi 2020). However, this analysis differs as it not only

tests this relationship but uses stepwise regression techniques to measure the significance of COVID-19 in the presence of several other market drivers, as they are added to the predictive market model. Ashraf (2020) includes several stationary control variables in his regression analysis, measuring democratic accountability, investment freedom and G.D.P. However, this analysis provides a more robust set of results on this contemporary matter, using a selection of continuous and temporal control variables. This analysis controls for several measures of market risk, including investor sentiment, volatility, economic health, monetary policy, counterparty credit risk, liquidity risk, safe-haven asset demand and the price of oil.

Secondly, this study adds to the literature published in the earlier stages of the pandemic by encompassing the entire 'curve' shape often referred to in epidemiology. Figure 1 illustrates this curve, showing the number of new COVID-19 deaths per day in each region analyzed. Capturing data over this entire period facilitates a robust analysis into the reaction of stock markets as growth in the virus began, surged, peaked, and eventually slowed. Thirdly, several recent publications on COVID-19 (see Liu et al. 2020; Harjoto et al. 2020; Ramelli and Wagner 2020) utilize an event study methodology to assess the market's reaction. However, as Oler et al. (2008) states, this methodology assumes the market's initial reaction to events reflects their true economic impact in an unbiased manner. This assumption may not hold in some cases, and when an event has complex ramifications that are difficult to predict, the initial market response may be biased or incomplete. Recent research utilizing this methodology assumes the 'first case' or 'first death' from COVID-19 as the event date. However, both standard epidemiology and Figure 1 indicate that a pandemic is a sudden and fast-evolving process and not a singular, isolated event in time. The announcement of a first case does not provide enough information to the market as to the true impact of an evolving pandemic. Although still a credible approach, the event study methodology may be deemed sub-optimal for this pandemic impact analysis. Therefore, this study instead utilizes multiple stepwise linear regression models, encompassing the entire pandemic period thus far to assess whether COVID-19 was significantly associated with the price of major stock market indices.

A number of interesting results emerged from this study. Firstly, while total COVID-19 cases were, on average, highly correlated to index price, it was not statistically significant in explaining the price of the Chinese SSE 180 index. However, major indices in Spain, Italy, the U.K., and the U.S.A. were found to be negatively and significantly related to the total number of COVID-19 cases. Controlling for other market drivers, the daily total number of COVID-19 cases remained significantly associated with equity index prices, using stepwise variable

selection methods. This is in line with the current research on COVID-19, discussed in the following section. Amidst global oil market tensions between Saudi Arabia and Russia, the price of Brent Crude Oil remained positively and statistically significant for China, the U.K., the U.S.A., and World indices. The LIBOR-OIS Spread, acting as a barometer for the level of fear within global banking systems also emerged as statistically significant for Italy, Spain, and World indices. Interestingly, Gold failed to act as a safe-haven or 'flight to safety' asset during the downturn, with a positive coefficient observed with U.S.A. index prices. Overall, the findings suggest a significant and negative impact of COVID-19 on major stock market indices. The following section describes the current literature on the topic, ranging from regression methods in finance, the impact of previous viral outbreaks on stock markets and finally to the research currently emerging from the ongoing COVID-19 pandemic.

Figure 1: *Daily Number of New COVID-19 Deaths per Region*

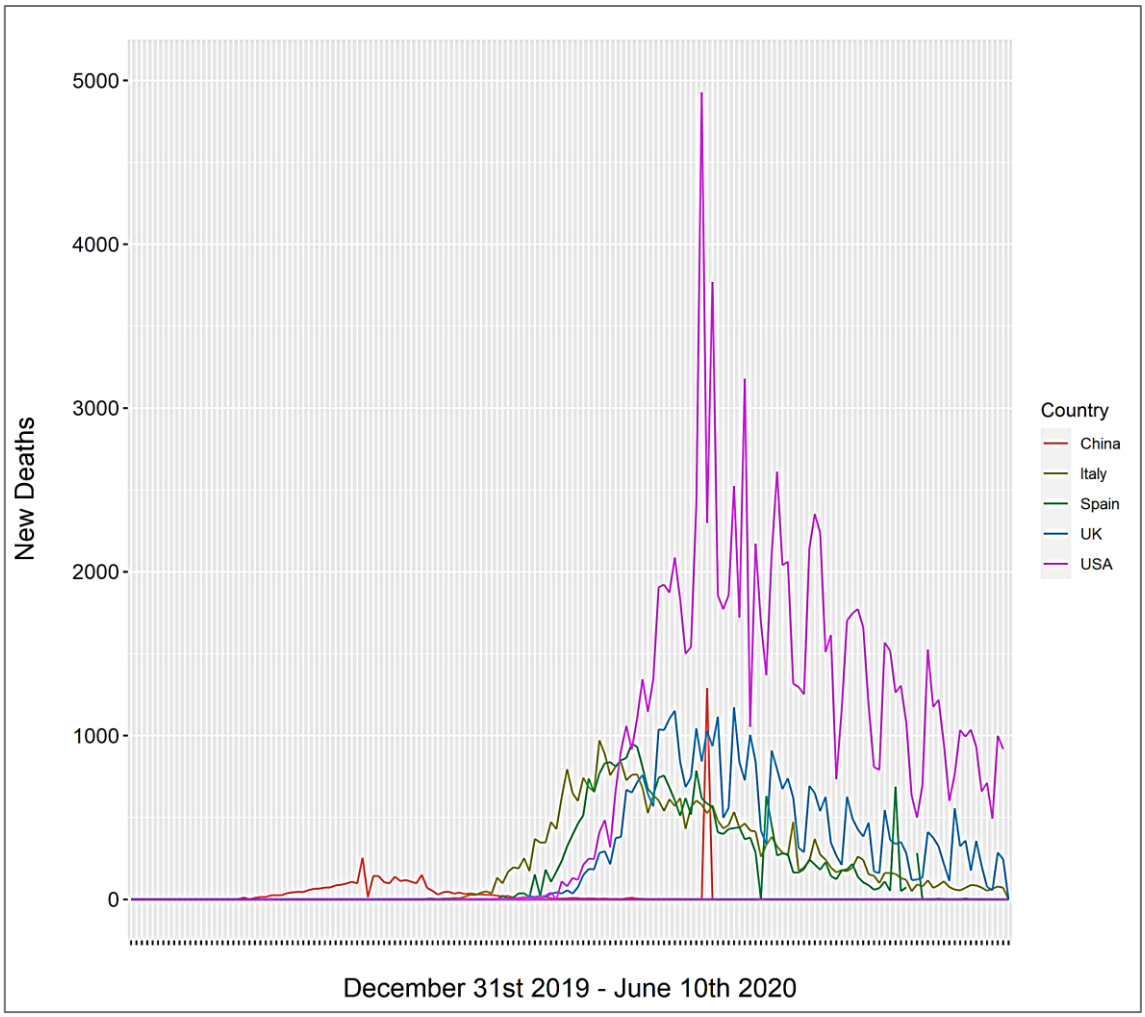


Figure 1: The daily number of new COVID-19 deaths in each country from December 31st, 2019 to June 10th, 2020. A roughly bell-shaped curve is observed for each country. Epidemics and pandemics tend to rise and fall in roughly bell-shaped curves, observed since (1840) by Farr. The high number of deaths in the U.S.A. skews the perception of the other curves. Individually plotted, a clear bell-shaped rise and fall is observed for all regions. Building on the literature published earlier in the pandemic, the data used in this study encompasses this entire bell-shaped curve.

1.3. Literature Review

Regression Analysis in Finance

Regression analysis has been used extensively to assess the impact of unprecedented shocks on local stock markets. Worthington (2008) conducted a long-term study spanning 1980-2003 on the impact of natural events and disasters on the Australian stock market. Utilizing auto-regressive techniques, the results indicated that natural events (including storms, floods, earthquakes, and bushfires) were not responsible for observed stock market volatility. Conversely, findings from Feria-Dominguez et al. (2017) indicate that natural disasters such as hurricanes directly contributed to significant abnormal returns on the NYSE. It could be argued that the sudden onset and shock of a global pandemic resembles that of a natural disaster. From the above studies, a potential explanation for the disparity in findings stems from the event window used and the concept of market efficiency. Siddikee et al. (2017) found that in their response to catastrophic disasters, large and organized exchanges such as the NYSE reacted abnormally and did not comply with the market efficiency rule. Taimur and Khan (2015) found that for short event windows of 1, 5, and 10 days, mean returns were not affected. However, 15-day long events windows did affect returns, conflicting with the semi-strong form of market efficiency. This finding is noteworthy as several recent studies on COVID-19 have utilized event study methods, despite the pandemic event lasting far longer than 1, 5, 10 or even 15 days. In favor of an event study methodology, this analysis makes use of stepwise variable selection and multiple linear regression analysis.

Multiple regression analysis, in its simplest form, attempts to model the linear relationship between independent and dependent variables. In other words, the main objective is to obtain a prediction for a dependent variable (Y), given the value(s) of 1 or '*n'* independent variables (X_1, $X_{2, \dots}$ X_n). Both Chan and Tse (1993) and Enke et al. (2011) use multiple regression models to identify variables significant at driving index returns. In their models, trading volume and the 3-month T-Bill rate proved significant in their forecasting abilities, respectively. Beisland (2011) highlights the importance and usefulness of financial ratios on stock returns using a multiple regression analysis. Cheng et al. (1990) asserts however, that the applicability of the multiple regression technique will always remain limited, due to its critical reliance on expectational factors. In other words, the assumption that historical relationships will exist in the future. Nonetheless, regression analysis remains an insightful methodology for a retrospective analysis on the impact of events on financial markets.

In this analysis, in order to identify the significant predictors of equity index prices, stepwise linear regressions are conducted. Using this technique, each independent variable is sequentially evaluated until the model contains only the variables deemed most contributory to the model's fit. Ou and Penman (1989) used stepwise logistic regression to construct an earnings prediction model from a large set of financial statement items, which was then used to dictate long and short positions in stocks. In the development of a hybrid machine learning prediction model, Enke and Mehdiyev (2013) also performed stepwise regression analysis to determine the combination of economic and financial indicators that had the strongest forecasting ability of index prices. Cheng (2015) followed the same methodology using 15 financial ratios in the development of a hybrid stock market prediction model. Therefore, the same methodology was chosen for this analysis in order to highlight whether the growth in COVID-19 remained statistically significant in the presence of other theoretical market drivers.

Impact of Past Viral Outbreaks on Stock Markets

The current literature on viral outbreaks is limited to an extent, as the scale of the current COVID-19 pandemic has been both unprecedented and unseen in over a century. Looking at the implications of the Spanish Flu of 1918-1919, there are a limited number of studies which examine the economic impacts. Among these studies, Bell and Lewis (2004) concluded a distinct lack of consensus among the findings. For example, Garrett (2007) found that cities with higher rates of influenza mortalities experienced higher real wage growth and higher income growth in the decade proceeding the pandemic. These findings echo those of Brainerd and Seigler (2003), who found the Spanish Flu led to an increase in U.S. economic growth in the 1920s. However, Garrett (2007) stresses that although the economic effects appeared short-term, no reasonable argument exists that suggests this outweighs the massive loss of life and economic activity. Conversely, findings from Barro and Ursua (2008) suggest that the Spanish Flu was perhaps the most important negative macroeconomic shock the world has experienced. In relation to the stock market, the Dow Jones Industrial Average Index was relatively unaffected over the three waves of the pandemic spanning 1918-1919 (Figure 2). However, war-time economics and post-war optimism are heavily confounding factors in any analysis of the stock market during the Spanish Flu, and so it does not provide a clear point of comparison for the present COVID-19 pandemic.

As the current outbreak unfolded, SARS was consistently referred to in the media as a template for how COVID-19 may impact the economy and stock markets (Avalos and Zakrajsek 2020; DeCambre 2020). However, despite a peak-to-trough stock price movement of 17.3% on the

S&P 500, Nippani and Washer (2006) declared SARS a 'non-event' for affected countries' stock markets overall, using an event-study regression analysis. Similarly, Loh (2006) found that the negative repercussions of SARS on airline stocks surfaced in the form of increased volatilities rather than lower mean returns. Conversely, Ming-Hsiang et al. (2007) analyzed Taiwanese hotel stock prices in the aftermath of SARS. Through a similar event-study regression, it was found that stock prices showed significant and abnormal negative returns on and after the day of the SARS outbreak. Despite a lack of consensus on the direct stock market implications, the overall financial cost of the outbreak was unequivocally significant, with DeLisle (2003) finding the losses to exceed that of the Asian financial crisis, estimating losses of $3 trillion in GDP and $2 trillion in financial market equity.

Figure 2: *Dow Jones Industrial Average Index: 1918 – 1919*

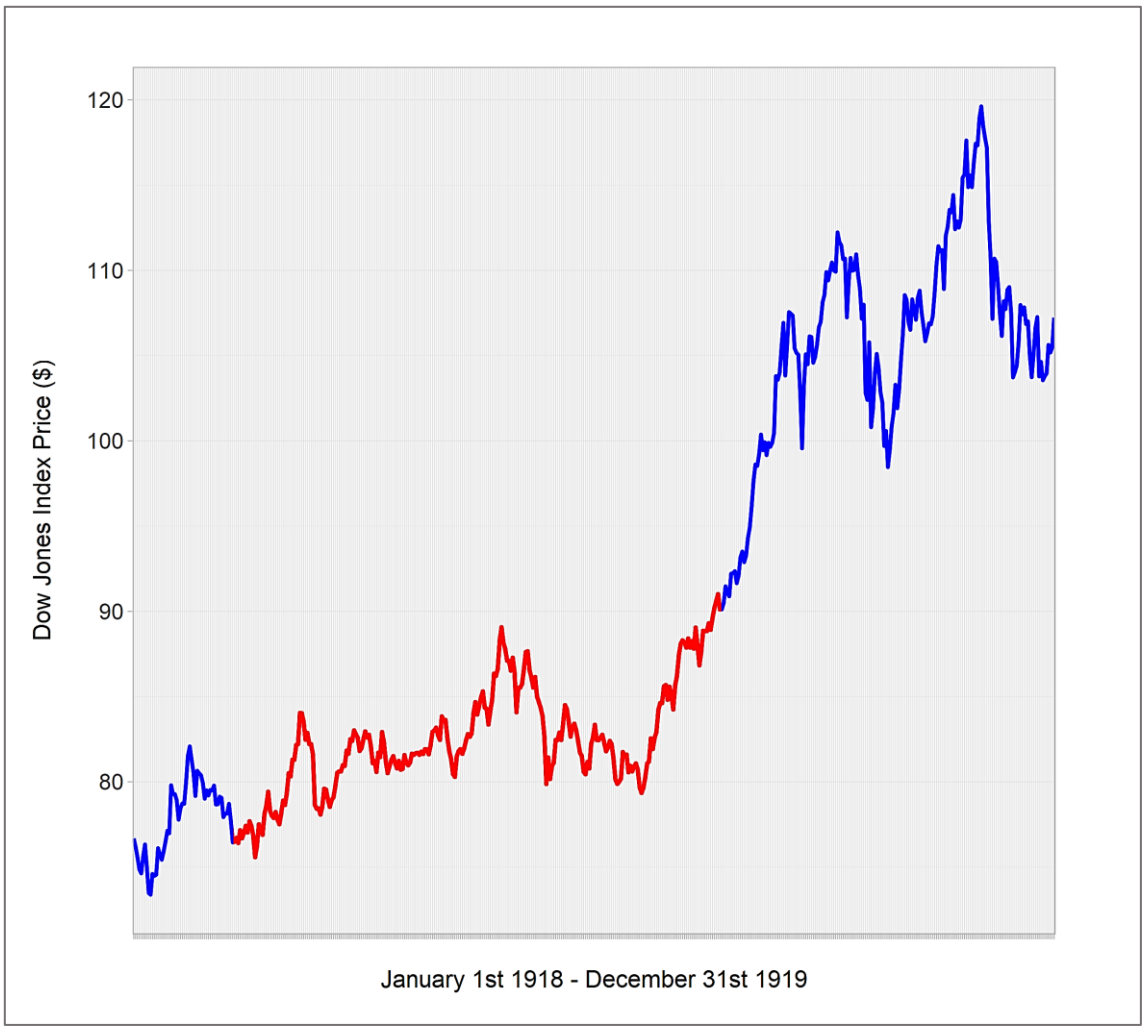


Figure 2: The daily price of the Dow Jones Industrial Average Index over the duration of the Spanish Flu pandemic. The red portion of the price path represents the time period encompassing all three waves of the pandemic. The end of the second (and worst) wave of the virus coincided with the end of WW1, prompting the evident price increase in the second half of the graph. Overall, the index did not appear significantly affected by the virus.

Ichev and Marinc (2018) and Del and Paltrinieri (2017) investigated the impact of the Ebola virus outbreak on U.S. and African financial markets, respectively. Ichev and Marinc (2018) observed a negative impact on U.S. listed stocks with business operations exposed to West Africa. Similarly, findings from Del and Paltrinieri (2017) suggest that the Ebola outbreak significantly affected the performance of mutual funds and returns on financial markets in Africa. Conversely, research examining the more recent Zika virus by Macciocchi et al. (2016) uncovered a minimal impact on stock markets. Following a linear regression methodology, the authors found that announcements from the WHO and World Bank were associated with occasional negative returns on the next day. However, these negative returns were not substantial nor sustaining, often recovering in the proceeding days. It is clear there is a lack of consensus regarding the direct impact of viral outbreaks on stock markets. Despite noteworthy economic and financial losses, evidence of a sustained and significant negative impact on stock markets is inconclusive from prior outbreaks.

Impact of COVID-19 Outbreak on Stock Markets

A number of studies have emerged in the first quarter of 2020 examining the impact of COVID-19 on financial markets. Among the methodologies used, it appears that panel regression analysis and event-study regressions are currently the favored methods. Panel data analysis is widely used in both epidemiology and econometrics, making it an appropriate methodology for the current COVID-19 pandemic. Using panel data spanning January to April, Ashraf (2020) found that stock markets reacted strongly with negative returns to growth in confirmed cases, but not to confirmed deaths. On the other hand, Ali et al. (2020) conducted bivariate regressions and found returns to be negatively and significantly related to COVID-19 deaths. Examining Chinese stock markets, Al-Awadhi et al. (2020) used a panel regression analysis and found that both the daily growth in confirmed cases and fatalities resulted in statistically significant negative returns on stock markets. Empirical results from both Zhang et al. (2020) and Albulescu (2020) indicate that COVID-19 directly impacted volatility on stock markets worldwide. Finally, Baker et al. (2020) used textual analysis to assess the unprecedented stock market impact of COVID-19. Analyzing newspaper articles published with relevant keywords, the authors' findings suggest no infectious disease outbreak as far back as 1900 has affected stock markets to the same extent as COVID-19. They found that news related to COVID-19 (both positive and negative) was the dominant driver of large daily U.S. stock market movements between 24th February 2020 through April 2020.

This analysis builds on the current COVID-19 related research by encompassing the entire pandemic period thus far, from 31st December 2019 to 10th June 2020. Using up-to-date data, this analysis examines the relationship between the growth in total COVID-19 cases and major stock market index prices. Using both forward selection and backward elimination of predictor variables, this analysis investigates whether the growth of COVID-19 in individual countries was significant at driving local index prices, while controlling for investor sentiment, volatility, economic health, monetary policy, counterparty credit risk, liquidity risk, safe-haven asset demand and the price of oil.

The study proceeds as follows: Section 2 describes the data sources and selection of variables for the regression models. Section 3 describes the empirical methodology used in the investigation. Section 4 presents the results, and Section 5 concludes the paper, including research considerations for the future as more COVID-19-related data becomes available.

2. Data

2.1. Data Collection

Daily data of cumulative case numbers and fatalities from COVID-19 was sourced from the European Centre for Disease Prevention and Control (ECDC). Specifically, data for China, Italy, Spain, U.K., U.S., and the World were obtained for this analysis, spanning the date range 31/12/2019 to 10/06/2020. These cumulative figures were subsequently transformed to provide eight daily indicators of COVID-19 that may have driven stock market indices:

Cases:	Deaths:
1. Total Number of Cases	5. Total Number of Deaths
2. Growth in Total Number of Cases	6. Growth in Total Number of Deaths
3. Number of New Cases	7. Number of New Deaths
4. Growth in Number of New Cases	8. Growth in Number of New Deaths

Growth rates were chosen as a simple daily arithmetic growth rate, in line with the reporting methods used by several media outlets throughout the pandemic (Harrison 2020; Cullen 2020; Bruce and James 2020). Preliminary testing of the eight COVID-19 indicators against index prices revealed 'Total Cases' as the highest correlated variable, on average across the countries. With these findings, it was decided to disregard the remaining 7 variables, since they were all either (i) highly correlated with Total Cases or (ii) not significantly correlated with index prices. Additionally, for the same date range, local equity index prices were obtained through the use

of a Bloomberg Terminal on the 10/06/2020. These indices are detailed below in Table 3. To concatenate trading days to calendar days, reported COVID-19 case numbers and deaths that occurred on weekend days were added to the next trading day's figures. Essentially, this was to tap into investor sentimentality and the reaction of stock market prices following an accumulation of news over the weekend period. As such, COVID-19 data was lagged one-day behind index prices. Figure 3 illustrates the daily new confirmed cases of COVID-19 alongside the daily price of the MSCI World Index, highlighting the sharp initial reaction of global markets to the onset of the pandemic, before rebounding and ultimately moving upwards alongside growth in the virus.

Table 3: *Equity Index Selection per Region*

Table 3: Presents the countries and equity indices chosen for analysis in this study. Beginning with China as the global origin of the virus, the remaining four countries were chosen as they chronologically established themselves as global epicenters of the virus over the course of the pandemic period.

Country	**Equity Index**	**Justification for Selection**
China	SSE 180	Global Origin of Virus
Italy	FTSE MIB	First European Epicenter
Spain	IBEX 35	Second European Epicenter
UK	FTSE 100	Final European Epicenter
USA	DJIA	Final Global Epicenter
World	MSCI World Index	World

The selection of explanatory and control variables detailed in Section 2.2 were obtained from various sources. Trading Volume, Volatility Indices, the price of Gold and the price of Brent Crude Oil were sourced from a Bloomberg Terminal. The Overnight Indexed Swap (O.I.S.) Rate, LIBOR-OIS Spread and T.E.D. Spread (or local alternatives) were sourced from the Federal Reserve Economic Database (F.R.E.D.), European Central Bank (E.C.B.), Bank of England (B.O.E.) and China Central Depository & Clearing (C.C.D.C.) as of the 10/06/2020.

Figure 3: *Daily MSCI World Index Price and Daily New Confirmed COVID-19 Cases*

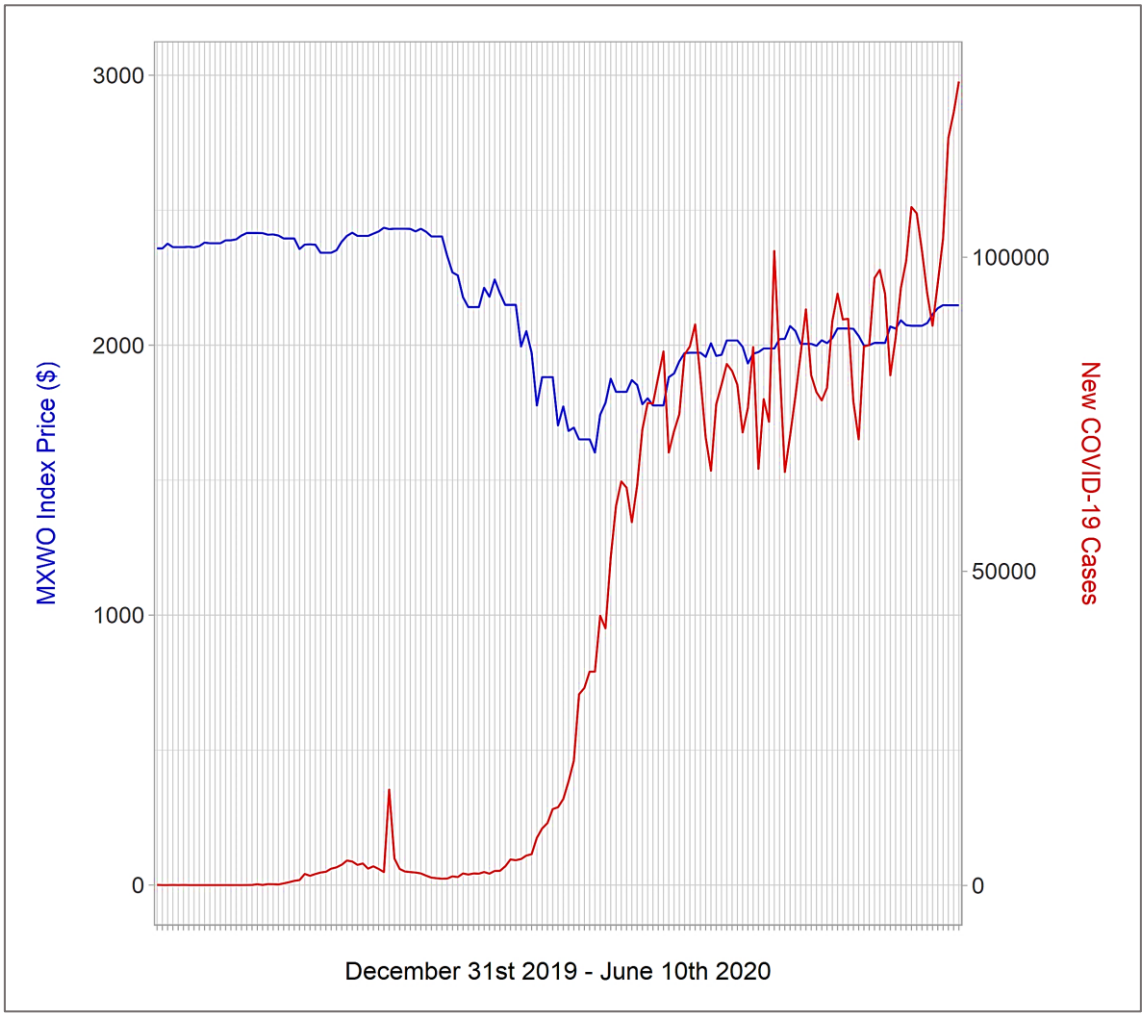


Figure 3: The daily price of the MSCI World Index plotted on a dual y-axis graph alongside the daily number of new confirmed COVID-19 cases. Initial growth in COVID-19 was met with a sharp price reaction of the World Index. However, this was short-lived, as the index began its recovery long before COVID-19 growth rates had peaked.

Table 4 presents summary statistics encompassing the growth of COVID-19 in all six regions included in this analysis. It can be seen that the maximum daily index return during the period was 11.37% (*USA – Dow Jones Index*) on the 29/02/2020 while the minimum daily return was observed as negative 16.92% (*Italy – FTSE MIB*) on the 21/02/2020. Summarizing COVID-19, the highest 1-day growth in total cases was 173.46% *(Italy - 22/02/2020)* and the highest 1-day growth in total fatalities was 460.00% (Spain - *10/03/2020*). It should be noted that large minimum and maximum figures were likely caused by noisy data in the early days of the pandemic, where numbers fluctuated up and down significantly. Of the five global epicenters of the virus, Spain saw the largest mean daily increase in cases / fatalities at 9.47% / 9.98% while China saw the lowest at 4.91% / 5.15%.

Table 4: *Summary Statistics: Index Returns & COVID-19 Growth*

Table 4: Presents summary statistics for both the stock market equity indices and the COVID-19 data used in this analysis. The daily logarithmic return was chosen for consistency across indices. In relation to the COVID-19 related growth rates, the simple arithmetic growth rate was used, in line with reporting methods used by several media outlets throughout the pandemic. All COVID-19 growth rates represent the 1-day growth rate in the cumulative number of cases or fatalities.

Country	Daily Growth in Variable:	Mean	Std. Dev.	Minimum	Maximum
China	Index Returns	-0.05%	1.40%	-7.63%	3.19%
	Total Cases	4.91%	12.77%	0.0012%	99.33%
	Total Fatalities	5.15%	13.67%	0.0216%	104.15%
Italy	Index Returns	-0.15%	2.86%	-16.92%	8.93%
	Total Cases	8.70%	30.35%	0.0757%	173.46%
	Total Fatalities	8.47%	24.72%	0.1527%	200.00%
Spain	Index Returns	-0.20%	2.65%	-14.06%	7.82%
	Total Cases	9.47%	17.65%	-0.3683%	109.86%
	Total Fatalities	9.98%	41.63%	-6.6708%	460.00%
U.K.	Index Returns	-0.16%	2.35%	-10.87%	9.05%
	Total Cases	9.00%	12.30%	-0.2112%	69.31%
	Total Fatalities	8.66%	22.46%	0.1357%	180.00%
U.S.A.	Index Returns	-0.06%	3.17%	-12.93%	11.37%
	Total Cases	8.85%	14.36%	0.9472%	91.63%
	Total Fatalities	9.16%	21.42%	0.4461%	200.00%
World	Index Returns	-0.08%	2.52%	-10.44%	8.41%
	Total Cases	8.85%	17.04%	0.6687%	163.86%
	Total Fatalities	9.94%	20.76%	0.2385%	183.33%

2.2. Variable Selection

In the preparation of any regression analysis, the selection of explanatory and control variables is of utmost importance. Therefore, drawing on previous financial literature, this analysis has selected a total of seven additional variables per region (in addition to the COVID-19 variable) to explain index prices. There was a limited number of predictor variables to consider due to the daily frequency of the COVID-19 data and the short timeframe of the current pandemic. As such, traditional economic indicators often used in literature could not be included due to the monthly or quarterly nature of their release. Considering this, the justification for the inclusion of the selected variables is discussed below. The specific rates and indices used for each region are detailed in Tables 6 through 9 in the Appendix.

Volatility Index: MEASURE OF: INVESTOR SENTIMENT | VOLATILITY

Volatility indices represent the 30-day forward expectation of volatility in financial markets. The index is typically derived from the price inputs of major index options. Although forward-looking, it is suitable as a control variable since it is an ex-post measure of market activity and the current sentiment of investors. Rosillio et al. (2014) assessed the effectiveness of implementing the V.I.X. as a Support Vector Machine feature in the prediction of the future price of the S&P 500 Index. The results showed that using the V.I.X. produced more accurate forecasts of the weekly change in price when compared to Buy and Hold strategies or models without the V.I.X. The study concluded by identifying the V.I.X. as a significant and influential addition to predictive trading systems, especially in bearish periods.

Overnight Indexed Swap (O.I.S.) Rate: MEASURE OF: MONETARY POLICY | ECONOMIC OUTLOOK

An overnight indexed swap (O.I.S.) is a derivative in which a fixed rate is paid in exchange for the daily accrual on a notional sum. (Huggins and Schaller 2013). The rate used to calculate this accrual is the O.I.S. rate. In the U.S.A, the Effective Federal Funds Rate is often quoted as this rate and is calculated as the volume-weighted median of actual overnight, unsecured transactions between banks. The O.I.S. rate therefore represents the cost of repeated overnight and unsecured lending from a central institution. The probability of considerable counterparty risks emerging overnight between such banks is considered relatively low. Additionally, an O.I.S. does not involve any initial cash flows. Therefore, it is essentially considered a riskless rate since credit risk will not be a major factor in its determination (Sengupta and Tam 2008).

By targeting this rate with liquidity provisions and monetary tools, a central institution can alter the cost of borrowing and affect the demand for goods and services. This, by proxy, may affect stock markets. In times of impending recessions, the rate can be decreased to stimulate loans, investments, and spending while the inverse can be carried out to cool an overheating economy. Results from Ekanayake et al. (2008) confirm this relationship, with stock markets typically decreasing following an increase in the Effective Federal Funds Rate. As such, it is a widely monitored economic indicator as it reveals the monetary policy stance of central banks and their future economic outlook. The indexed rate varies per region. Calculated in largely the same way as the Effective Federal Funds Rate, the EONIA, SONIA and SHIBOR Overnight Index are used in the Eurozone, U.K., and China, respectively.

LIBOR-OIS Spread: MEASURE OF: COUNTERPARTY CREDIT RISK | LIQUIDITY RISK

Unlike the O.I.S. rate, the LIBOR (referencing a cash instrument) may be affected by credit and liquidity risk under times of economic stress. This is analyzed in detail by Brunnermeier (2009) and Sarkar (2009) during the global financial crisis of 2007/08, when lenders became skeptical about total market solvency. This resulted in liquidity hoarding, a drying up of liquidity and ultimately a liquidity spiral. While a bank lending at the three-month LIBOR deposit rate is exposed to three months' worth of credit risk, a three-month overnight indexed swap exposes the lender to just one-day of credit risk at each sequence. Therefore, as conditions deteriorate in the banking sector, LIBOR rises to represent both credit and liquidity risks in the market while the O.I.S. rate will have significantly less exposure to these risk factors. The spread between the LIBOR and the O.I.S. rate can therefore be used as a barometer for fears of insolvency in the banking sector. Widening of the LIBOR-OIS spread indicates a deterioration of credit conditions, since banks that are normally seen as safe are now viewed with more uncertainty and counterparty risk. While LIBOR USD/GBP are used in the U.S.A. and the U.K., the EURIBOR and SHIBOR are used in the Eurozone and China, respectively.

T.E.D. Spread: MEASURE OF: COUNTERPARTY CREDIT RISK | LIQUIDITY RISK

The T.E.D. Spread is the difference between the yield on a three-month U.S. T-Bill and the three-month LIBOR USD Rate. This difference represents the risk premium charged on top-rated interbank loans versus risk-free loans, and so is often seen as an indicator of the perceived credit risk in an economy (Boudt et al. 2017; Taylor and Williams 2009). Drobetz (2000) found that the T.E.D. Spread is affected by three major factors: (i) world political stability, (ii) balance of trade, and (iii) fiscal policy in the United States. When political uncertainty is high and the balance of trade is decreasing, the spread should widen. Therefore, it can also be used as an indicator of overall economic health. Regressing the spread against rates of industrial production, Drobetz (2000) confirmed that the yield differential should be higher during economic recessions, when investors are seeking safe assets. As such, it is a suitable indicator for credit risk and overall economic risk during the COVID-19 pandemic. For countries other than the U.S., the yield on the generic 3-month government bond and the respective interbank offered rate was used in this study.

Gold: MEASURE OF: SAFE-HAVEN ASSET DEMAND

Gold has been described as both a hedge and a safe haven in differing market conditions, by Baur and Lucey (2010). While a safe haven asset loses none of its value in the event of a market

crash, a hedge is one that on average, does not co-move with the market. The authors found both definitions to be true of gold, with results indicating that (i) Gold is a hedge against stocks, (ii) Gold is a safe haven in extreme stock market conditions and (iii) Gold is a safe haven for stocks for 15 trading days after an extreme shock. As such, it is a suitable control variable for the recent shock of COVID-19 on financial markets and the extended period of economic uncertainty that transpired. If gold is a significant independent variable, it may indicate that stock market prices were depressed as a result of investors flocking out of traditional investments and into safe haven assets and hedges.

Oil: MEASURE OF: INFLATION | GLOBAL OIL PRICE TENSIONS

Conventional wisdom holds that like gold, oil too has an inverse relationship with stock markets. As oil prices rise, the cost of energy increases, resulting in systemic inflation in the economy as the overall cost of doing business rises. However, the literature is divided regarding the strength of this relationship. Cong et al. (2008) found that shocks to the price of oil did not have a statistically significant impact on real stock returns of major Chinese indices. Conversely, Kling (1985) and Sadorsky (1999) found that increases in the price of crude oil were associated with stock market declines. Analyzing the issue more specifically, Kilian and Park (2009) identified that the reaction of U.S. stock markets to oil price shocks differed greatly depending on whether the shock was driven by demand or supply elements in the oil market. They concluded that production shocks were far less significant to stock markets when compared to shocks to the global aggregate demand for oil. In light of the highly volatile global oil markets during March 2020, see Masson and Winter (2020), the price of oil is a relevant and potentially significant factor in recent stock market prices.

Trading Volume: MEASURE OF: MARKET ACTIVITY

As a direct indicator of market activity, trading volume was included to assess whether fluctuations in the previous day's trading volume played a role in driving index prices the proceeding day. The conventional wisdom of technical analysts that volume leads price movements is not universally observed in empirical findings. However, findings from the likes of Chan and Tse (1993) show an implicit positive correlation between price and volume, confirming the usefulness of incorporating volume data in forecasts of future price.

Guided by relevant financial literature, the above variables have been chosen to represent the many dynamics underlying financial markets. By including measures of investor sentiment, volatility, economic health, monetary policy, counterparty credit risk, liquidity risk, safe-haven

asset demand and the price of oil, this analysis seeks to identify which, if any of the above were significant in driving index prices during the COVID-19 pandemic. It is both evident and intuitive that COVID-19 is linked to the recent downturn in financial markets. Utilizing stepwise regression methods and comparing both forward selection and backward elimination of potential predictor variables, this analysis investigates the significance of raw COVID-19 data, while controlling for the above factors.

3. Methodology

3.1. Stepwise Multiple Regression Analysis

With eight potential predictors identified, there are 2^8 potential model combinations that could be used to explain index prices in each country, a phenomenon known as combinatorial explosion. Since it would not be feasible to manually test all 256 combinations, a method of variable selection known as stepwise regression has been used to identify the most significant variables in the optimal multiple regression model. This procedure consists of the iterative addition and deletion of variables, one at a time, according to a specific information criterion. This criterion, discussed in Section 3.2, can be regarded as a measure of the model's performance or fit. Variables will be added or deleted until this criterion decides the model will achieve no significant improvement from the addition of more variables. Stepwise regression typically consists of three steps:

1. Forward Selection: This technique starts off with a minimal model and no predictor variables. The dependent variable (index price) will be explained by '1'. In other words, it is an intercept only model. Variables will then be added one at a time until either all variables are in, or until the performance of the model would decrease as a result of more variables.

2. Backward Selection: This technique begins with a model in which all variables are included in the model. At each step, the variable which would cause the largest decrease in the model's information criterion will be deleted. In other words, the model will iteratively remove the least contributory variable, and stops when all variables are statistically significant.

3. Stepwise Selection: Therefore, combining both forward and backward selection, the model used in this analysis begins as a simple, intercept only model explaining index price. It then sequentially adds the most contributory variable. At each step, variables which no longer contribute significantly to the model are removed. The process continues in this manner until the lowest information criterion has been achieved.

The ‘R’ statistical language within RStudio (R Core Team 2013) in addition to the MASS Package (Ripley et al. 2020) was used in this analysis, the working code of which can be found in the Appendix. With the chosen measurement of COVID-19 growth (Total Cases) and the seven additional market drivers outlined in Section 2.2, stepwise regressions combining both forward and backward steps were conducted, the results of which are discussed in Section 4.

3.2. Penalized-Likelihood Criteria: A.I.C. & B.I.C.

At the core of the stepwise regression technique is a criterion for model and variable selection within the complete set of available variables. As Halsey (2019) states, the p-value has long been the figurehead of statistical analysis. However, recently, it has been criticized for the limited information it provides about data. In particular, model building based on p-values may lead to an inflated risk of type 1 errors (Mundry and Nunn 2008). P-values can also be misleading where predictor variables may be collinear. In this case, high p-values would be observed for variables which are not necessarily useless. For this analysis, two widely used criteria, the Akaike Information Criterion (A.I.C.) from Akaike’s (1974) study and the Bayesian Information Criterion (B.I.C.) from Schwarz’s (1978) analysis were considered.

A.I.C. is a goodness of fit measure which favors smaller residual errors while penalizing for excessive predictor variables, thus helping to curtail overfitting. It is an estimate of a constant plus the relative distance between the unknown true likelihood function of the data and the fitted likelihood function of the entire model. A lower A.I.C. value thus implies a model is closer to the true model. B.I.C. is largely similar. However, the B.I.C. favors a more parsimonious model, in contrast to the AIC. It is an estimate of a function of the posterior probability for a model being true, under a Bayesian setup. Like the A.I.C., the lower the B.I.C., the more the model is considered to be true. The formulae for both criterion and the penalization terms (highlighted in red) are shown below, illustrating B.I.C.’s higher penalization of excessive model parameters.

$$\boldsymbol{AIC = \mathrm{n}.\, ln(SSE) - \mathrm{n}.\ln(\mathrm{n}) + 2(k+1)}$$

$$\boldsymbol{BIC = \mathrm{n}.\, ln(SSE) - \mathrm{n}.\, ln(\mathrm{n}) + ln(n)(k+1)}$$

Where:

- n: Number of Observations
- S.S.E: Error Sum of Squares
- k: Number of Predictors
- k + 1: Number of Predictors + Intercept

If the likelihood gain from adding a parameter is small, a stepwise regression under B.I.C. may select a different model to AIC. This effect is dependent on sample size, however, evident from the penalization terms above. Initial stepwise regressions revealed almost identical models from both criteria, with some marginal differences. From Shmueli's (2010) study "To Explain or Predict", it is stated that the A.I.C. may be better suited for prediction purposes since it favors a greater closeness of fit, while it can be argued that the B.I.C. is better suited for the purposes of explanation, as it is consistent. As such, to ensure consistency and in favor of a parsimonious model, the B.I.C. results were chosen for analysis in the following section.

4. Empirical Results & Discussion

Table 5 below presents the results of the six stepwise linear regression models. Beginning with China, the models are presented in chronological order as the virus established global epicenters, moving from China (1), to Italy (2), Spain (3), the U.K. (4) and finally to the U.S.A. (5) which, at the time of writing, has established itself as the global epicenter of COVID-19. Additionally, a model including global COVID-19 growth and the MSCI World Index (6) was included. In the following section, results will first be presented on an index level. Following this, each significant independent variable will be discussed, beginning with the variable found in the most models, and progressing to the least-included variable.

4.1. Results – Index Models

China (1):

The final stepwise regression model under the Bayesian Information Criterion (B.I.C.) concluded that of the seven potential independent variables, the price of Brent Crude Oil was the most contributory independent variable, and so was added first to the model. This was followed by the Volatility Index, the TED Spread and Trading Volume, respectively. The following model to explain the price of the SSE 180 Index was produced:

$$SSE\ 180\ Index = 8394.707 + 9.73 * Brent - 16.01 * Volatility - 187.76 * TED\ Spread + 2.24e^{-8} * Volume + \varepsilon$$

Italy (2):

Like China, Brent was considered the most explanatory variable and was added to the model first. This was followed by the iterative addition of the Volatility Index, Total Cases of COVID-19 and the LIBOR-OIS Spread. Following the addition of the LIBOR-OIS Spread, the B.I.C.

considered Brent to be insignificant, subsequently removing it from the model. The following model to explain the price of the FTSE MIB Index was produced:

$$FTSE\ MIB\ Index = 26399.42 - 129.22 * Volatility - 0.01 * Total\ Cases - 9{,}556.77 * LIBOR.OIS + \varepsilon$$

Spain (3):

For Spain, the same stepwise addition and deletion pattern as Italy was observed. Beginning with Brent, the Volatility Index, Total Cases of COVID-19 and the LIBOR-OIS Spread were sequentially added to the model. Following the addition of the LIBOR-OIS Spread, the model's fit subsequently improved as a result of the deletion of Brent. The following model to explain the price of the IBEX-35 Index was therefore produced:

$$IBEX.35\ Index = 10492.73 - 49.57 * Volatility - 0.01 * Total\ Cases - 2953.01 * LIBOR.OIS + \varepsilon$$

U.K. (5):

Once more, the stepwise regression under B.I.C. first added Brent to the model, followed by the Volatility Index, Gold and Total Cases of COVID-19, respectively. The following model to explain the price of the FTSE-100 Index was produced:

$$FTSE\ 100\ Index = 8939.5 + 20.15 * Brent - 27.65 * Volatility - 1.47 * Gold - 0.001 * Total\ Cases + \varepsilon$$

U.S.A. (5):

Unlike the previous four models, Volatility emerged as the most contributory independent variable for the Dow Jones Index. This was followed by Brent, Gold and Total Cases of COVID-19, respectively. The following model to explain the price of the Dow Jones Industrial Average Index was produced:

$$Dow\ Jones\ Index = 16521.74 - 121.52 * Volatility + 68.05 * Brent + 6.35 * Gold - 0.001 * TotalCases + \varepsilon$$

World (6):

Like the USA, Volatility was first added to the World Index stepwise regression model. This was followed by the iterative addition of Brent and the LIBOR-OIS Spread, respectively. The following model to explain the price of the MSCI World Index was produced:

$$MSCI\ World\ Index = 2207.59 - 7.66 * Volatility + 4.95 * Brent - 68.53 * LIBOR.OIS + \varepsilon$$

Table 5: *Stepwise Regression Models Explaining Equity Index Price in Each Region*

Models (1) to (6) represent the final stepwise regression model for each region under the Bayesian Information Criterion. Significant independent variables are listed on the left-hand-side. The dependent variable (Equity Index Price) is listed from left-to right, beginning with the Chinese SSE 180 index. A number without parentheses represents a regression coefficient while a number inside parentheses represents a standard error.

	Dependent Variable: Equity Index Price					
	SSE 180 CHINA	**FTSE MIB** ITALY	**IBEX 35** SPAIN	**FTSE 100** UK	**DJIA** USA	**MSCI World** WORLD
	(1)	(2)	(3)	(4)	(5)	(6)
Independent Variable:						
Volatility	-16.009***	-129.223***	-49.574***	-27.648***	-121.519***	-7.658***
(Standard Errors)	(2.237)	(4.264)	(1.675)	(1.728)	(12.321)	(0.567)
Total Cases		- 0.011***	- 0.006***	- 0.001***	- 0.001**	
(Standard Errors)		(0.001)	(0.001)	(0.0002)	(0.0002)	
Brent	9.730***			20.151***	68.048***	4.952***
(Standard Errors)	(1.940)			(2.477)	(15.221)	(0.583)
LIBOR-OIS		-9,556.768***	-2,953.014***			-68.534***
(Standard Errors)		(1,896.521)	(772.988)			(22.724)
Gold				- 1.469***	6.347***	
(Standard Errors)				(0.392)	(1.985)	
TED Spread	-187.761***					
(Standard Errors)	(61.699)					
Volume	$2.24e^{-8}$***					
(Standard Errors)	($7.53e^{-9}$)					
Constant	8,394.707***	26,399.420***	10,492.740***	8,939.500***	16,521.740***	2,207.593***
(Standard Errors)	(113.995)	(173.608)	(67.642)	(720.989)	(3,583.432)	(38.766)
Observations	117	117	117	117	117	117
R^2	0.781	0.950	0.954	0.966	0.917	0.932
Adjusted R^2	**0.774**	**0.949**	**0.953**	**0.965**	**0.914**	**0.930**
Res.Std.Error	156.644 (df = 112)	754.005 (df = 113)	292.494 (df = 113)	155.112 (df = 112)	853.126 (df = 112)	59.418 (df = 113)
F Statistic	100.116*** (df = 4;112)	718.500*** (df = 3;113)	788.786*** (df = 3;113)	807.094*** (df = 4;112)	308.733*** (df = 4;112)	518.420*** (df = 3;113)

Note: *p<0.10, **p<0.05, ***p<0.01

4.2. Discussion – Significant Variables

Volatility Index:

Volatility indices remained as a significant predictor for each of the regions examined in this analysis. In line with Rosillo et al. (2014), the inclusion of a volatility index produced better fitting models in each instance. This was unsurprising, as measures of implied volatility have historically seen an inverse relationship with stock prices. As Giot (2005) notes, measures of implied volatility typically perform well in challenging market conditions characterized by bear prices and high volatility, the same market conditions observed during COVID-19.

Total Cases of COVID-19:

The number of total COVID-19 cases remained a significant independent variable in four of the six regression models. Namely, for the Italy, Spain, U.K., and U.S.A. models. It should be noted that new research currently in pre-publication by He at al. (2020) suggests that based on cremation estimates, there was a significant under-and-delayed reporting of the COVID-19 epidemic in China. As such, access to more accurate data may have revealed total cases as significant in the case of China, also. In each instance, a negative coefficient was observed for the COVID-19 variable, indicating that as the total number of cases rose, the index price decreased. These findings are in line with those currently available from the ongoing pandemic. Ashraf (2020), Al-Awadhi et al. (2020) and Sansa (2020) all reported similar findings, with COVID-19 cases negatively and significantly associated with stock markets in each study.

No association between COVID-19 and the MSCI World Index was observed, using global case data. The index represents large and mid-cap companies in 23 developed markets. Therefore, it is possible that the impact of the virus was diversified over the different local peaks and troughs in both COVID-19 growth (evident in Figure 1) and stock market growth, across the affected countries. Heston and Rouwenhorst (1994) found that diversification across countries (within one industry) was a much more effective tool for risk reduction than diversification across industries (within one country). Although global indices may provide the benefits of international diversification, the findings in this study suggest that in times of serious crisis and economic disruption, this is not the case. Although the world index did not appear significantly associated with COVID-19, it was not immune from the concomitant impact of widening LIBOR-OIS spreads and decreasing oil prices, discussed below.

Brent Crude Oil:

Amid high tensions in the global oil market, it was unsurprising that the price of Brent Crude Oil remained significant in four of the six regression models. It was interesting to note the inclusion of Brent for China, the U.K., U.S.A., and the World as a whole, but not for Spain and Italy in the final models. With this, the conventional inverse relationship between oil and stock prices is not present in these findings either. Each coefficient is positive, indicating that increases or decreases in the price of oil is associated with the same directional movement in index prices. However, in the case of COVID-19, this directional relationship makes sense. An unprecedented erosion of oil demand as a result of pandemic-induced economic shutdowns initially drove oil prices down, signaling a diminished level of total economic activity to financial markets. Alongside a depressed demand, a breakdown in negotiations between OPEC, Russia and Saudi Arabia resulted in both Saudi Arabia and Russia announcing unprecedented plans to increase oil production. As demand continued to diminish due to COVID-19, an additional surplus in supply led to an oil price war between Russia and Saudi Arabia with Brent decreasing as much as 26% in early March alone (Egan 2020).

In normal conditions, the price of oil may be used as a gauge for inflation, with increases in price signaling an overall increase in the cost of doing business, which typically leads to decreases in stock prices (Sadorsky 1999). However, under current COVID-19 conditions, the price of oil may instead act as a gauge of economic activity and of global political tensions, thus explaining the positive correlation observed with index prices. As the price of oil rose from May onwards, so too did global index prices. Kilian and Park address this relationship in their (2009) study, stating that the resilience of stock markets in the presence of increasing oil prices can be explained by strong global demand for industrial commodities, which can more widely be representative of increasing economic activity. As such, any indication of increasing economic activity during COVID-19 was a positive indicator to financial markets.

LIBOR-OIS Spread:

The LIBOR-OIS Spread remained statistically significant at the 99% level in three of the six models, while statistically significant at the 95% level for the U.K. An increase in the spread for both Italy and Spain (the difference between the 3M EURIBOR and EONIA Rates) was associated with a decrease in the respective index prices. Similarly, an increase in the USD spread (the difference between the 3M LIBOR USD and the Effective Federal Funds Rate) was associated with a decrease in the MSCI World Index price. Notable credit and liquidity risks

were present within major economies before COVID-19 took hold. Before the pandemic, the I.M.F. warned in (2019) that 40% of corporate debt in major economies could be at risk in a global downturn. As COVID-19 took hold and economies shutdown, corporate liquidity effectively dried up. Companies began drawing on short-term credit lines in order to shore up balance sheets, with over $120 trillion drawn by European and American companies in less than one month (Platt et al. 2020). As pandemic-induced shutdowns took hold, consumer demand diminished, corporations drew increasingly on short-term credit lines, and the demand for liquid assets increased significantly (Bank of England 2020).

As a result, the LIBOR-OIS spread rose to the highest level seen since the global financial crisis of 2007/08. Detailed in Section 2.2, widening of this spread represents increasing credit risks and fears of insolvency within the banking sector. A report by Duren and Saucier (2020) stated that a persistent ascent could result in higher borrowing costs, rising interest rates and ultimately a hindering of global economic expansion in the wake of COVID-19. Therefore, the significance of the spread for Spain and Italy, but not for the United States was intriguing. One possible explanation is rooted in the response of central institutions to the pandemic. The Federal Reserve notably responded swiftly, reducing the Federal Funds Rate by 50bp in an attempt to promote investment and a greater level of economic activity. In the same announcement, daily overnight secured loans up to at least $175 billion were provided, alleviating money market pressures and ensuring an ample supply of reserves (Federal Reserve 2020). This injection of confidence to the market importantly came just before tensions in the global oil markets emerged.

On the other hand, the E.C.B. lagged briefly in their response (Collins and Gagnon 2020). The difference in policy response may explain why the LIBOR-OIS remained significant in European markets, but not American. These findings are similar to those of Ramelli and Wagner (2020), who found that corporate debt and cash holdings emerged as important value drivers as the virus spread to Europe and the U.S.A.

Gold:

Mixed findings were observed for Gold, with an increase in Gold prices associated with a decrease in the U.K.'s FTSE 100 index. Meanwhile, a positive coefficient for model (5) indicated that as the price of Gold prices decreased, so too did the DJIA index. This is evident from Gold's peak-to-trough losses of 14.22% concomitant with the decline of the DJIA. Although gold is often regarded as an 'international flight to safety asset' (Akyildirim et al.

2020), the evidence from this analysis suggests otherwise, and is consistent with recent findings by Corbet et al. (2020). As in this analysis, no significant relationship between Gold and Chinese stock markets was found during the current pandemic. It can be suggested that in times of serious financial and economic disruption, gold fails to act as a safe-haven and may instead act as an amplifier of financial contagion.

T.E.D. Spread:

The T.E.D. Spread remained significant with a negative coefficient in model (1) only, indicating that an increase in the perceived credit risk within the Chinese economy resulted in a decrease in the price of the SSE 180 index. While the T.E.D. measures the risk associated with government-to-bank lending, the LIBOR-OIS represents the risk of bank-to-secured lending facilities. While the LIBOR-OIS was significant in three regions, the T.E.D. was only significant in one. This was likely due to the high level of correlation between both variables, which saw the U.S. T.E.D. Spread and LIBOR-OIS correlated by 98% over the period analyzed. Conversely, China's equivalent measure of the T.E.D. Spread (3M SHIBOR minus 3M ChinaBond Yield) saw a lesser 64% correlation, potentially explaining its inclusion in model (1) but not in any other model. However, both indicators provide a gauge of overall credit risk and economic uncertainty. It is clear that increased risks within the global banking system, through both the T.E.D. Spread and the LIBOR-OIS Spread, were associated with decreasing stock market prices during the COVID-19 pandemic period.

Volume:

Finally, trading volume was significant only in model (1), with a positive coefficient of 7.530×10^{-9} observed. While volume has been found to be positively correlated with stock prices in normal conditions (Chan and Tse 1993), it is arguable that during times of significant economic and financial disruption there are more significant drivers of index price. While the COVID-19 crisis appeared on paper to have effectively abated in China by the beginning of March, the other regions examined were just beginning to experience the pandemic. As such, it is unsurprising to see a different set of predictors at play within Chinese markets. As European and American markets were beginning to price in the possibility of an economic shutdown, Chinese markets were already quantifying the impact of an 'under-control' pandemic.

The following section concludes the study, with a summary of the findings and potential limitations, alongside recommendations for future impact studies on the COVID-19 pandemic.

5. Conclusion

This study examined the stock market response to the ongoing COVID-19 pandemic, investigating and identifying potential drivers of equity index prices. Using the daily total count of confirmed COVID-19 cases and a selection of seven additional economic and financial indicators, the findings from this analysis suggest that major index prices responded negatively and significantly to increases in the total number COVID-19 cases. Additionally, measures of Implied Volatility, the price of Brent Crude Oil and the LIBOR-OIS Spread emerged as significant drivers of index prices. Pandemic-induced economic shutdowns arguably drove financial market volatility, while business closures and travel restrictions eroded the demand for oil. As businesses shut, counterparty credit and liquidity risks emerged, driving the LIBOR-OIS Spread, before central institutions ultimately intervened with monetary policy responses. As such, while Total Cases of COVID-19 emerged as negatively and significantly associated with index prices, it can be argued that overall, COVID-19 has infected all corners of the financial sector. These findings concur and add to the current collection of COVID-19 literature, confirming the significant reaction of stock markets to the recent pandemic.

By introducing additional market factors and utilizing stepwise regression techniques, this analysis builds on the current literature by investigating whether growth in COVID-19 remained statistically significant as other financial indicators were introduced to the market model. Unlike previous studies, this analysis employs seven robust measures of market activity, volatility, monetary policy, counterparty credit risk, liquidity risk, safe-haven asset demand and the price of oil. The number of economic and financial indicators chosen must be noted as a potential limitation of this study. Due to the monthly and quarterly release of traditional economic and financial indicators, common measurements of productivity, consumer spending, inflation and G.D.P. could not be incorporated in this daily analysis. Moving forward, as the pandemic becomes abundantly more quantifiable, a larger pool of features can be incorporated into a COVID-19 financial impact analysis. With this, previous major viral outbreaks can also be examined in the same way and compared to COVID-19, in order to identify the significance of the same market forces under various pandemic conditions. Additionally, as in this study, stepwise regression techniques incorporating COVID-19 data may be used in future work alongside machine learning techniques in the development of stock price forecasting models, see Tsai et al. (2018). Although the initial stock market reaction appears short-lived just as in previous pandemics, this study robustly confirms the significant and negative impact of the COVID-19 pandemic on major stock market index prices.

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7. Appendix

7.1 Additional Tables

Table 6: *Volatility Indices*

Country	Index
China	CBOE VXFXI China ETF Volatility Index
Italy	EURO STOXX 50 Index
Spain	EURO STOXX 50 Index
UK	EURO STOXX 50 Index
USA	CBOE DJIA Volatility Index
World	CBOE Volatility Index (VIX)

Table 7: *Overnight Indexed Swap Rates*

Country	Index
China	SHIBOR Fixing Overnight Index Rate
Italy	Euro Overnight Index Average (EONIA)
Spain	Euro Overnight Index Average (EONIA)
UK	Sterling Overnight Index Average (SONIA)
USA	Effective Federal Funds Rate
World	Effective Federal Funds Rate

Table 8: *Interbank Offered Rates*

Country	Index
China	Shanghai Interbank Offered Rate (SHIBOR)
Italy	Euro Interbank Offered Rate (EURIBOR)
Spain	Euro Interbank Offered Rate (EURIBOR)
UK	London Interbank Offered Rate (LIBOR GBP)
USA	London Interbank Offered Rate (LIBOR USD)
World	London Interbank Offered Rate (LIBOR USD)

Table 9: *Other Rates*

Country	Index
3M Bond	Yield on 3M Generic Government Bond
Gold	Brent Crude Oil (C01) Bloomberg
Oil	Spot Price of Gold (XAU) Bloomberg

7.2 R Code

```
#CHINA#

#Import China
china <- read.csv("chinalag.csv", stringsAsFactors=FALSE)
china <- as.data.frame(china);

ChinaAll = lm(SSE ~ TotalCases + Volume + VXFXI + SHIBOROIS + TEDCN + Gold +
Brent, data = china)

#China Forward Selection Model

ChinaFit = lm(SSE ~ 1, data=china)

stepAIC(ChinaFit, scope=formula(ChinaAll), direction = "both", k = log(117))

## Call:
## lm(formula = SSE ~ Brent + VXFXI + TEDCN + Volume, data = china)
##
## Coefficients:
## (Intercept)        Brent        VXFXI        TEDCN       Volume
##   8.395e+03    9.730e+00   -1.601e+01   -1.878e+02    2.246e-08



#ITALY#

#Import Italy
italy <- read.csv("italylag.csv", stringsAsFactors=FALSE)
italy <- as.data.frame(italy);

ItalyAll = lm(FTSEMIB ~ TotalCases + Volume + V2X + LOISEUR + TEDIT + Gold + Brent
, data = italy)

#Italy Forward Selection Model

ItalyFit = lm(FTSEMIB ~ 1, data=italy)

stepAIC(ItalyFit, direction = "both", scope=formula(ItalyAll), k = log(117))

## Call:
## lm(formula = FTSEMIB ~ V2X + TotalCases + LOISEUR, data = italy)
##
## Coefficients:
## (Intercept)          V2X   TotalCases      LOISEUR
##   2.640e+04   -1.292e+02   -1.122e-02   -9.557e+03



#SPAIN#

#Import Spain
spain <- read.csv("spainlag.csv", stringsAsFactors=FALSE)
spain <- as.data.frame(spain);

SpainAll = lm(IBEX ~ TotalCases + Volume + V2X + LOISEUR + TEDSP + Gold + Brent, d
ata = spain)
```

```r
#Spain Forward Selection Model

SpainFit = lm(IBEX ~ 1, data=spain)

stepAIC(SpainFit, direction = "both", scope=formula(SpainAll), k=log(117))

## Call:
## lm(formula = IBEX ~ V2X + TotalCases + LOISEUR, data = spain)
##
## Coefficients:
## (Intercept)          V2X   TotalCases      LOISEUR
##   1.049e+04   -4.957e+01   -5.750e-03   -2.953e+03



#UK#

#Import UK
uk <- read.csv("uklag.csv", stringsAsFactors=FALSE)
uk <- as.data.frame(uk);

UKAll = lm(UKX ~ TotalCases + Volume + V2X + LOISUK + TEDUK + Gold + Brent, data =
uk)

#UK Forward Selection Model

UKFit = lm(UKX ~ 1, data=uk)

stepAIC(UKFit, direction = "both", scope=formula(UKAll), k =log(117))

## Call:
## lm(formula = UKX ~ Brent + V2X + Gold + TotalCases, data = uk)
##
## Coefficients:
## (Intercept)        Brent          V2X         Gold   TotalCases
##   8.939e+03    2.015e+01   -2.765e+01   -1.469e+00   -6.603e-04



#USA#

#Import USD
usd <- read.csv("usdlag.csv", stringsAsFactors=FALSE)
usd <- as.data.frame(usd);

USDAll = lm(DJIA ~ TotalCases + Volume + VXD + LOISFF + TEDUS + Gold + Brent, data
= usd)

#USD Forward Selection Model

USDFit = lm(DJIA ~ 1, data=usd)

stepAIC(USDFit, direction = "both", scope=formula(USDAll), k =log(117))



## Call:
## lm(formula = DJIA ~ VXD + Brent + Gold + TotalCases, data = usd)
##
## Coefficients:
## (Intercept)          VXD        Brent         Gold   TotalCases
##   1.652e+04   -1.215e+02    6.805e+01    6.347e+00   -5.217e-04
```

```
#WORLD#

#Import World
world <- read.csv("worldlag.csv", stringsAsFactors=FALSE)
world <- as.data.frame(world);

WorldAll = lm(MXWO ~ TotalDeaths + Volume + VIX + LOISFF + TED + Gold + Brent, dat
a = world)

#World Forward Selection Model
WorldFit = lm(MXWO ~ 1, data=world)

step(WorldFit, direction = "both", scope=formula(WorldAll), k=log(117))

## Call:
## lm(formula = MXWO ~ VIX + Brent + LOISFF, data = world)
##
## Coefficients:
## (Intercept)          VIX        Brent       LOISFF
##    2207.593       -7.658        4.952      -68.534
```

```
#RESULTS#

CN <- lm(SSE ~ Brent + VXFXI + TEDCN + Volume, data = china)
IT <- lm(FTSEMIB ~ V2X + TotalCases + LOISEUR, data =italy)
SP <- lm(IBEX ~ V2X + TotalCases + LOISEUR, data =spain)
UK <- lm(UKX ~ Brent + V2X + Gold + TotalCases, data= uk)
US <- lm(DJIA ~ VXD + Brent + Gold + TotalCases, data = usd)
WD <- lm(MXWO ~ VIX + Brent + LOISFF, data = world)

stargazer(CN, IT, SP, UK, US, WD, title="Results", align=TRUE, out="all.htm")
```